

upGrad

Pre-read: the data-engineering mental model on AWS

Prepare for M01

Why read this before the module?

Come prepared to

- distinguish storage, compute, cataloguing and querying responsibilities;
- recognise the trade-off between exploratory convenience and reproducible pipelines; and
- bring one data source from your work context to map onto the architecture.

Four responsibilities, not four interchangeable tools

Responsibility	Typical AWS service	Question to ask
Store source data	Amazon S3	What is immutable, and what is the retention policy?
Discover schema	AWS Glue Data Catalog	Which table definition is trusted by downstream users?
Transform at scale	AWS Glue / EMR	Can the job be rerun from versioned inputs?
Query interactively	Amazon Athena	Who can query which prefix, and at what cost?

Reading prompt

Question – Map one data flow

Choose a source you know: product events, orders or support tickets. Where would raw objects land? What creates a curated table? Which team owns the schema? Write down one uncertainty to bring to Session 01.

Key takeaway

Key takeaway – Design for the next question

A useful data platform preserves raw evidence, makes curated data discoverable and lets transformations be rerun when the business question changes.